Aditya Akbar

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PROFILE

Computer Science undergraduate at Universitas Brawijaya with hands-on experience in machine learning, computer vision, NLP/LLM applications, time series data modeling, and applied AI product workflows. Built end-to-end projects across biometric security, medical imaging, air quality forecasting, Indonesian NLP, and privacy-aware AI companions. Strong in data preprocessing, experimentation, model evaluation, technical documentation, and clean Python-based implementation.

Relevant AI domains: Time Series, computer vision, NLP, LLM applications.

EDUCATION

Bachelor of Computer Science in Brawijaya University - GPA 3.73 2024 - Ongoing

SKILLS

Programming: Python, SQL, Java, PHP, HTML/CSS/JavaScript
ML & Data Science: Pandas, NumPy, Scikit-learn, SHAP, Captum, Matplotlib, Tableau
Deep Learning & CV: PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, MONAI, Ultralytics, Mediapipe
NLP & LLM: Indonesian text preprocessing, RAG, FAISS, PostgreSQL, Gemini/OpenAI-based workflows, prompt orchestration
Tools & Engineering: Git, Anaconda, Flask, Laravel, Docker basics, Azure Functions, technical documentation

PROJECT EXPERIENCE

DSN-MUI UMKM RAG | RAG, BGE-M3, BM25, RRF, Qwen3-VL, FastAPI, ngrok

- Built a modular RAG prototype for Indonesian Islamic finance guidance, enabling MSMEs to query DSN-MUI fatwas, akad provisions, and supporting dalil from PDF-based sources.
- Designed a precomputed retrieval pipeline with PDF extraction, Arabic OCR support, chunking, BGE-M3 embeddings, BM25 indexing, hybrid retrieval, RRF fusion, and statistical reranking.
- Integrated optional Qwen3-VL visual reasoning to extract Arabic dalil from document crops and improve source-grounded answer generation.
- Exposed the system as a FastAPI backend and shared it through ngrok for public demo access, API testing, and frontend/backend integration.

Leksara | Python, NLP, Indonesian Text Preprocessing

- Developed an open-source Python library for cleaning Indonesian e-commerce text, including slang, abbreviations, punctuation, stopwords, contractions, and noisy review content.
- Improved dataset readiness for downstream NLP and machine learning tasks through reusable preprocessing workflows.

Senandika / AI MoodBand | LLM Orchestration, Privacy-Aware Memory, PostgreSQL + FAISS

- Designed a non-clinical emotional companion workflow with listen-first prompting, non-diagnostic guardrails, schema-validated outputs, and safety-aware response flows.
- Implemented privacy-preserving vector memory and stress/emotion analytics to support trustworthy AI behavior in sensitive user scenarios.

AirSafe School | Time-Series Forecasting, LightGBM, SHAP, Azure Functions, LLM Recommendations

- Built an end-to-end air quality forecasting and school recommendation system using hourly PM2.5 data, weather integration, feature engineering, and spatial interpolation.
- Used LightGBM and SHAP to generate explainable forecasts, then connected prediction outputs to LLM-powered recommendations for practical decision support.

Face Spoofing Detection | Computer Vision, DINOv3, ConvNeXt, FSFM, Ensemble Learning

- Developed a six-class face anti-spoofing classification pipeline for real and spoofed faces, including printed, screen, mask, mannequin, and unknown spoof categories.
- Implemented dual-view processing using full-image context and face-crop detail, with embedding-based deduplication, anomaly review, test-time augmentation, and ensemble fusion.
- Achieved around 0.979 macro F1 validation performance with DINOv3-based modeling and documented preprocessing, model comparison, threshold logic, and error analysis for research-style reporting

3D Brain Tumor Segmentation | PyTorch, MONAI, MedicalNet-ResNet18, U-Net, U-Net++

- Built a 3D multi-modal MRI segmentation pipeline using T1w, T1ce, T2w, and FLAIR data to support healthcare AI research.
- Designed controlled architecture and training experiments, achieving Dice mean around 0.85-0.87, with WT around 0.88-0.90, TC around 0.82-0.86, and ET around 0.75-0.81.
- Delivered reproducible code, architecture diagrams, experiment documentation, and research-paper-ready explanations.

PROFESSIONAL & RESEARCH EXPERIENCE

Research Member & Research Intern | Intelligent Systems Laboratory, FILKOM Universitas Brawijaya 2025 - Ongoing

- Conduct AI research involving deep learning experimentation, data preparation, model training, evaluation, and reporting for image segmentation-focused projects.
- Document methods, experimental results, limitations, and architecture decisions for internal reports and research outputs.

Data Scientist | BCC FILKOM Universitas Brawijaya 2025 - Ongoing

- Contribute to applied AI and data science initiatives, including project-based machine learning development, data analysis, and technical exploration.
- Support department programs while strengthening practical skills in experimentation, feature engineering, and AI product workflows.

Machine Learning Mentor | GDGoC on Campus Universitas Brawijaya 2025 - Ongoing

- Mentor students in machine learning learning programs and study jams, covering data preprocessing, visualization, ML workflows, and introductory computer vision.
- Prepare learning materials and hands-on exercises to help participants understand practical AI development.

Data Science Intern | BCC - Airbnb Optimum Price Feb 2025 - Mar 2025

- Processed and analyzed Airbnb listing data using Python to identify pricing patterns and key features affecting listing value.
- Built a machine learning model for optimum price prediction, covering preprocessing, EDA, feature selection, model training, and evaluation.

ACHIEVEMENT

- **Finalist - Data Analytics Competition at Find IT! 2026, Universitas Gadjah Mada: Face Spoofing Detection for biometric security classification.**
- **Top 7 Finalist - Datavidia Arkavidia 10.0, Institut Teknologi Bandung: multivariate time-series forecasting for ISPU category prediction.**
- **12th Place - Machine Learning Competition Data Slayer 3.0, Telkom University: drowsiness detection using video-based machine learning.**

LEADERSHIP & ORGANIZATION EXPERIENCE

Vice Head of Talent Development Division | BCC FILKOM Universitas Brawijaya 2025 - Ongoing

- Lead member development initiatives by shaping learning roadmaps, coordinating upskilling activities, and monitoring talent growth within the student technology community.

Project Officer - Data Science Mini Bootcamp 2025 | BCC FILKOM Universitas Brawijaya 2025 - Ongoing

- Managed end-to-end execution of a multi-day data science bootcamp, including timeline planning, curriculum flow, mentor/speaker coordination, participant experience, and post-event documentation.

Vice Project Officer - BCC Open Recruitment 2026 | BCC FILKOM Universitas Brawijaya 2025 - Ongoing

- Coordinated recruitment planning across six departments by aligning selection flow, timeline management, cross-division communication, and stakeholder follow-up to keep the process consistent.